



▲ SHIP'S NATURALIST

The main object of the 19th-century voyages was to discover and chart new lands, along with any hazards to navigation. But they also carried scientists on board—most famously Charles Darwin, who joined the 1831–36 voyage of HMS *Beagle*, shown here anchored off Tierra del Fuego. While at sea, Darwin collected marine organisms, examined the nature of ocean water, and speculated on the origins of volcanic islands, coral reefs, and atolls, and the species that lived on them.



▲ THE CHALLENGER EXPEDITION

The 1872–76 voyage of HMS *Challenger*—an ex-survey ship—was the first attempt to understand how oceans work. The physicists, chemists, and biologists on board measured and sampled everything possible by dredging, trawling, and taking soundings using long cables—even in the deepest parts of the ocean. In the process they discovered more than 4,700 new species and gained a rough picture of the ocean floor with its ridges, chasms, and abyssal plains.



▲ MODERN OCEANOGRAPHIC RESEARCH

Modern research ships gather data on oceans throughout the world, feeding it back to oceanographic institutes such as Woods Hole and Scripps in the US, and at Southampton and Naples in Europe. These institutions draw together the many sciences involved in oceanography, including oceanic physics and chemistry, marine biology, geology, and meteorology. The complex interaction of all these disciplines makes oceanography one of the most demanding sciences.

OCEANOGRAPHY

Oceanographic research was pioneered during the voyages of Captain James Cook (1728–1779) and Charles Darwin (1809–1882), who gathered valuable information on ocean currents, temperatures, and depths, as well as oceanic geology and marine life. But the real science of oceanography began with the voyage of HMS *Challenger* in the late 19th century. This voyage was planned with the sole object of gathering as much data about the oceans as possible. It formed the foundation of further research, all coordinated by land-based oceanographic institutions.

